

2023



Training the Next Generation of Builders at Queensborough Community College

Learn how Queensborough Community College uses AWS Academy curriculum to prepare students for careers in cloud computing.

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150+

AWS Certifications obtained by QCC st

~ 4.000

Built a pipeline

of diverse early career cloud talent

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Companies worldwide are increasingly adopting the cloud, and the demand for professionals with cloud-computing skills is rising. However, this rising demand only widens the cloud-skills gap. There are not enough job candidates who have the cloud skills that businesses need. To bridge this gap and help learners from all backgrounds prepare for careers in cloud computing, [Queensborough Community College \(QCC\)](#) offers its students cloud-skills training designed by Amazon Web Services (AWS) experts.

[AWS Academy](#) provides higher education institutions with a free, ready-to-teach cloud-computing curriculum that prepares students to pursue industry-recognized certifications and in-demand cloud jobs. The curriculum helps educators stay at the forefront of AWS Cloud innovation so that they can equip students with cloud skills. Since first adopting AWS Academy in 2019, almost 4,000 QCC students have completed AWS Academy courses.



Opportunity | Closing the Cloud-Skills Gap by Preparing a Diverse Student Body for In-Demand Roles

QCC is one of the 25 campuses of the City University of New York (CUNY) public university system—and one of the highest-rated community colleges in New York State. Given its location in the most diverse county in the United States, the college attracts students from all walks of life. Many students hail from communities that are underrepresented in the technology sector, are adult learners returning to school after long breaks, have jobs or are raising families alongside their college classes, and require financial aid. QCC further supports diversity by proudly serving as both a minority-serving institution and a Hispanic-serving institution. In the 2021–2022 academic year, 31 percent of QCC students were Black while 29 percent were Hispanic. As a result, QCC trains diverse pools of learners on the practical skills that they need to be successful.

Prof. Michael Lawrence, a faculty member in the QCC Engineering Technology Department since 2014, saw the cloud-skills gap as an opportunity to take advantage of AWS Education Programs. At that time, he began using [AWS Educate](#) to supplement the courses that he taught. AWS Educate offers hundreds of hours of free, self-paced training and resources for new-to-cloud learners—including hands-on labs in the AWS Management Console.

In October 2018, Prof. Lawrence learned about the innovative cloud courses within AWS Academy that are developed and maintained by AWS experts and are designed to fit seamlessly within an educator's existing course structure.

Solution | QCC Adopts the Ready-to-Teach AWS Academy Cloud Computing Curriculum

QCC adopted AWS Academy courses, both as part of its community college curriculum and as free, grant funded, continuing education and workforce development classes. To teach the courses, Prof. Lawrence attended several AWS classes and earned several AWS Certifications. Five other instructors in the QCC Engineering Technology Department have earned AWS Certifications and are also teaching AWS Academy courses.

Instructors can use the predesigned AWS Academy curriculum to teach cloud skills and their application in the workplace. AWS Academy designs its courses with input from industry experts and customers, and AWS experts keep the courses up-to-date with current technology. Plus, there is room for instructors to add their own spin. Prof. Lawrence found that he could easily integrate discussions beyond the cloud into the courses he teaches using the AWS Academy curriculum. "The AWS Academy material provides the what, while I talk about why that what is important," he says.

The AWS Academy courses include videos, quizzes, handbooks, and hands-on labs to give students a foundation upon which later lessons build. The videos often feature talks by industry influencers, like the former chief technology officer of Lyft, who discusses how students would practice coding by recreating the architecture that Lyft was built on. The hands-on lab exercises feature real-life scenarios to teach important elements of cloud computing, such as using containers and automatic scaling. "I wouldn't be doing this without AWS Academy," says Prof. Lawrence. "I wouldn't have had the time to put this all together."

AWS Academy courses also help prepare students to earn [AWS Certifications](#), which validate technical skills and cloud expertise.

Outcome | Building a Pipeline of Diverse, Cloud-Skilled Talent

Since adopting AWS Academy, Prof. Lawrence estimates that he has taught almost 4,000 students cloud computing, with approximately 150 AWS Certifications earned. The most common certification earned is the [AWS Certified Cloud Practitioner](#), which demonstrates a foundational understanding of AWS Cloud concepts. Some students have earned more advanced AWS Certifications, including [AWS Certified SysOps Administrator – Associate](#), which demonstrates experience in managing workloads on AWS, or [AWS Certified Solutions Architect – Associate](#), which demonstrates knowledge and skills in AWS technology.



Prof. Lawrence is especially inspired by the students who take advantage of the evening continuing education classes at QCC. “These are free courses where students have no skin in the game other than their personal desire,” he says. “And these students get most of the AWS Certifications.”

Given these results, QCC plans to continue using the AWS Academy curriculum as it helps to build a pipeline of talent that local employers can tap into to close their cloud-skills gaps.

“Don’t let AWS Academy go away!” Prof. Lawrence says. “Folks at AWS are doing something incredible and wonderful, which is teaching students vital skills to get them better employment prospects instead of being stuck in dead-end jobs. It’s like helping them get out of a burning building. These courses give them the skills that they can use to really do something.”

About Queensborough Community College

Queensborough Community College is part of the City University of New York (CUNY) and located in one of the most diverse urban areas in the United States. It is a minority-serving institution, with many students working while taking classes.

AWS Services Used

AWS Academy

AWS Academy provides higher education institutions with a free, ready-to-teach cloud computing curriculum that prepares students to pursue industry-recognized certifications and in-demand cloud jobs.

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AWS Educate

Build your cloud skills at your own pace, on your own time, and completely for free.

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AWS Certified Cloud Practitioner

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